

Requirements for "Allround scoring system"

1. Scope

- 1.1 The system provides multi-event athletics scoring for decathlon/heptathlon.
- 1.2 Two user interfaces are provided: a desktop UI and a Web UI. Both use the same core logic.

2. Core Domain Requirements

2.1 Scoring

- 2.1.1 The system shall compute points per event using the 2001 IAAF scoring rules.
- 2.1.2 Input results shall be validated against reasonable bounds before scoring.
- 2.1.3 Rounding/point totals shall follow the IAAF convention (integer points).

2.2 Competitors

- 2.2.1 The system shall manage up to 40 competitors.
- 2.2.2 For each competitor, the system shall store name, per-event raw results, per-event points, total points and result position.
- 2.2.3 Competitor data shall be editable (e.g., correct a result).

2.3 Standings

- 2.3.1 The system shall provide current standings at any time, showing per-event points, total points and current result position.
- 2.3.2 Standings shall update when new or edited results are saved.

2.4 Save & Resume

- 2.4.1 The system shall allow saving the current state (competitors, results, points, standings).
- 2.4.2 The system shall allow resuming from a previously saved state.
- 2.4.3 Saved state shall be UI-agnostic: a state saved in one UI can be resumed in the other.

2.5 Export

- 2.5.1 The system shall export results to an Excel-compatible file containing: competitor name, event results, points per event, total points and result position.
- 2.5.2 The export filename shall be unique (e.g., include a timestamp) to avoid overwriting previous files.

2.6 Architecture and Modularity

- 2.6.1 Scoring rules, validation, and data structures shall reside in a shared core module independent of any UI framework.
- 2.6.2 Both UIs shall depend on the shared core and must not duplicate scoring logic.

3. Interface Requirements (High-Level)

3.1 Common Capabilities

- 3.1.1 Both UIs shall support: adding competitors, entering results, viewing standings, saving/resuming, and exporting results.
- 3.1.2 Given identical inputs, both UIs shall produce identical scores and totals.

3.2 Desktop UI

- 3.2.1 Shall present clear feedback for invalid inputs and operation errors.

3.3 Web UI

- 3.3.1 Shall be accessible locally in a modern desktop browser.
- 3.3.2 Should be suitable for automated UI testing (e.g., stable labels/structure), without prescribing specific selectors.

4. Data and Validation

- 4.1 The system shall define acceptable input ranges per event (time/distance units as appropriate) and reject out-of-range values.
- 4.2 The system shall handle invalid or missing input with user-readable error messages.
- 4.3 Units used for input and display shall be consistent across UIs.

5. Quality Attributes

- 5.1 Usability: Key tasks (add competitor, enter result, view standings, save, export) shall be achievable with minimal steps.
- 5.2 Performance: Operations shall remain responsive with up to 40 competitors and all supported events.
- 5.3 Reliability: Save/resume shall not corrupt data; failed loads shall provide a recoverable error message.

6. Constraints and Out of Scope

- 6.1 Offline/local operation only; multi-user/network features are out of scope.
- 6.2 Detailed UI styling, specific control layouts, and test selector naming are out of scope.